

Operating Limits & Setpoints – Quick Reference Sheet

Nexum Suum | Boiler System Benchmark Chart

<u>Parameter</u>	<u>Typical Operating Range</u>	<u>Notes / Visual Guidance</u>
Boiler Pressure	12–150 PSI	Low-pressure: 12–30 PSI; High-pressure: 100–150 PSI
Stack Temperature	275–450°F (gas); 350–550°F (oil)	<300°F: corrosion risk; >550°F: heat loss
Cycles per Hour	3–6 (gas); 2–5 (oil/steam)	High cycling reduces efficiency
NOWL	Mid-sight glass (±1") Also where steam and water meet the pressure vessel	Should bounce slightly; avoid dropping below
Blowdown Frequency	Daily (LWCO/Sight) and Bottom ; Weekly (Mud Drums Water Tube Boilers)	
Feedwater Temp	180–220°F	Avoids thermal shock; improves efficiency
Burner Flame Color	Blue/yellow stable	Yellow tips = incomplete combustion
Stack O ₂	3–6% (gas); 4–7% (oil)	Low O ₂ = CO risk; High O ₂ = excess air
Combustion Efficiency	78–85% (basic); 85–95% (condensing) based on stack test	

Visual Setpoint Zones (Color Key)

Zone	Description	Action
Green	Optimal range	Maintain & record
Yellow	Monitor—approaching limit	Schedule maintenance review
Red	Out of range—correct immediately	Investigate root cause ASAP

How to Use: Post this sheet in boiler room. Use colored tags/magnets to mark live readings. Log alerts on the Efficiency Scorecard.